

WEEK 4 – POWER BI REPORT

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INTRODUCTION

This report presents the Week 4 analysis of Shopify stock data in Power BI. The dashboard was enhanced with calculations and interactive features to explore historical price behaviour and changes in trading activity. Moving averages were used alongside visual reports to make trends easier to interpret.

DATA PREPARATION

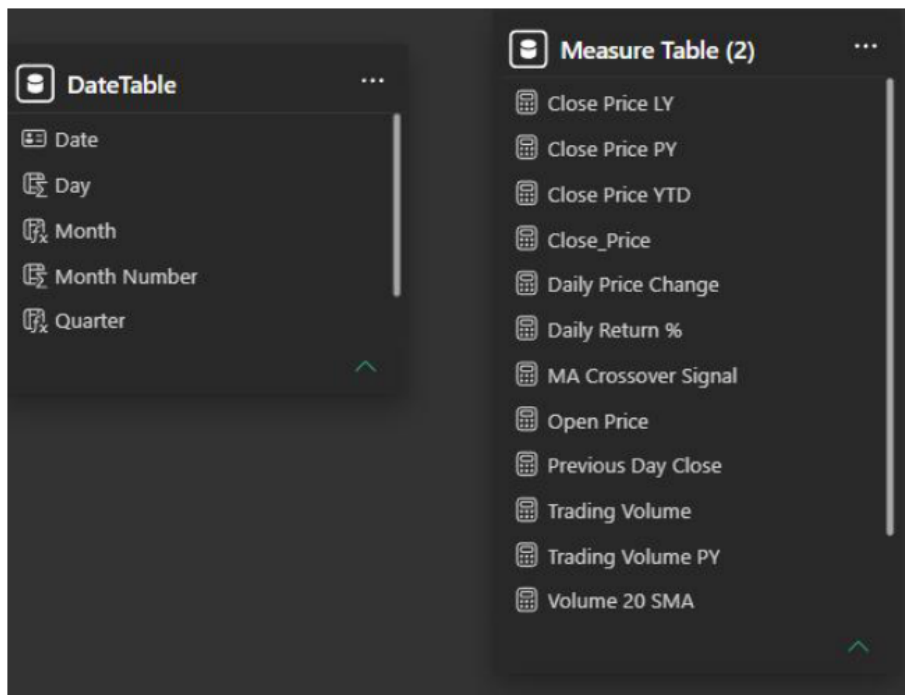
The Shopify dataset was brought into Power BI and arranged for time-based analysis. The date field supports viewing the information across different periods, while the prepared data is used in charts and DAX calculations to examine prices and trading volume.

DAX MEASURES

DAX measures were added for the 20-Day SMA, 50-Day SMA, and Moving Average Crossover Signal. These calculations help compare recent and longer-term movements and provide a clearer view of possible changes in the stock trend.

TRADING VOLUME ANALYSIS

Trading volume was reviewed to see how the level of market activity varied over time. The 20-Day and 50-Day moving averages were used to compare shorter-term changes with the broader pattern. This view helps highlight periods when trading activity was relatively high or low.

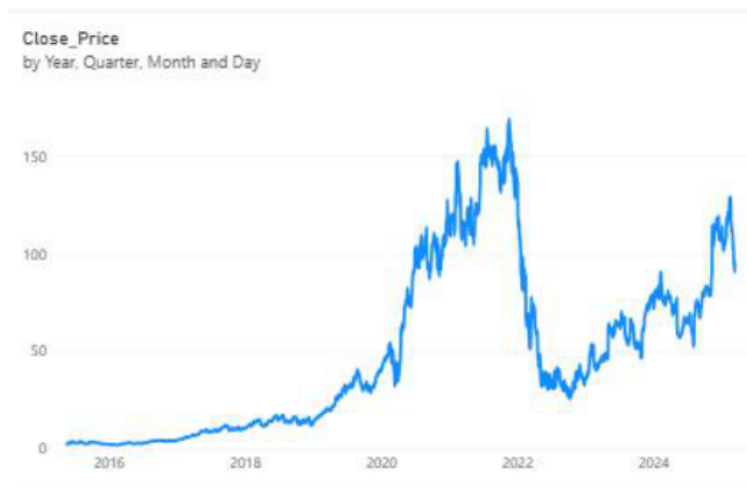


```
1 Trading Volume =  
2 SUM('SHOP_2015-05-21_2025-03-16'[Vo
```

PRICE TREND ANALYSIS

Shopify's closing-price history from 2015 to 2025 was examined to observe how its value changed over the years. The Daily Return % measure adds a day-to-day comparison, making short-term price fluctuations easier to identify alongside the overall trend.

```
Volume 50 SMA =  
VAR CurrentDate = MAX('DateTable'[Date])  
VAR PeriodDays = 50  
VAR DateInPeriod =  
    DATESINPERIOD(  
        'DateTable'[Date],  
        CurrentDate,  
        -PeriodDays,  
        DAY  
    )  
RETURN  
    CALCULATE(  
        AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]),  
        DateInPeriod  
    )
```



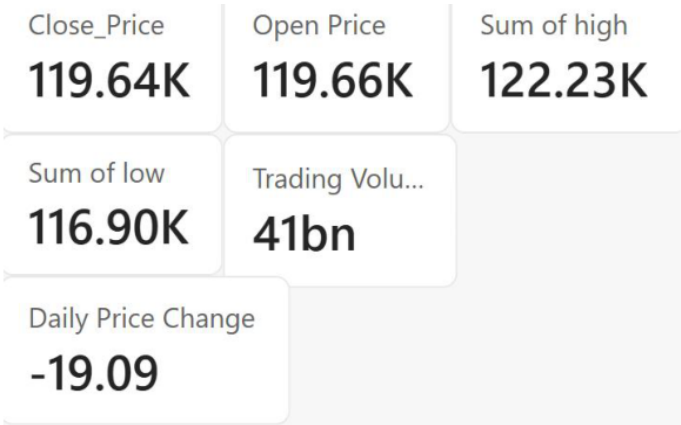
MONTH SLICER AND DYNAMIC FILTERING

A Month slicer was included so users can narrow the dashboard to selected months. Choosing a month helps focus the analysis on the relevant time period and makes it easier to compare stock prices and trading volume without changing the main report layout.



CUSTOM STOCK TOOLTIP

A dedicated Stock_Tooltip page was set up to reveal key values when the pointer is placed over a chart. The tooltip contains Open Price, High Price, Low Price, Close Price, Trading Volume, and Daily Change %. It provides additional detail on demand while keeping the primary dashboard uncluttered.



CONCLUSION

The Week 4 work strengthened the Shopify stock dashboard with DAX-based measures, moving-average analysis, and views of price and volume patterns. Month-based filtering gives users control over the period they explore, while the custom tooltip offers extra stock details when needed. Together, these additions make the report more interactive and support clearer exploration of stock performance.